

Constraint-Conditioned Proportional Switching-Time Control for a Three-Cell Switched-Affine DC–DC Converter

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Abstract

Switched-affine power converters can be regulated by moving their switching instants instead of by commanding a continuous input amplitude. This paper develops a low-complexity cycle-to-cycle controller for that setting. A first-order model maps the cycle-boundary state error and the switching-instant offsets to the error one cycle later, and a discrete linear-quadratic design supplies a static feedback gain. Unconstrained feedback can request infeasible schedules, so a scalar conditioning step contracts the entire offset vector until every dwell duration meets its lower bound. The method is simulated on a three-cell multilevel DC–DC converter with eight operating modes and a $3\mu\text{s}$ minimum dwell duration. From a large initial perturbation, the normalized cycle-to-cycle error falls from 0.9854 to 0.0086 in 80 cycles, a 99.1% reduction, while the shortest observed dwell duration stays at $3.0\mu\text{s}$. Under the local model the raw-action admissible polyhedron is positively invariant, and the simulated trajectory enters it after 3 conditioned cycles. Outside that polyhedron the conditioned loop is a convex combination of the open-loop and unconditioned closed-loop matrices, which yields a common-Lyapunov certificate that conditioning cannot destabilize the local model whenever the nominal orbit is open-loop stable. The controller is therefore an inexpensive constrained baseline for periodic-orbit stabilization, subject to its local-model and uniform-scaling limitations.

Keywords: switched-affine systems; periodic-orbit stabilization; proportional state feedback; switching-time control; dwell-time constraints; DC–DC converter.

1 Introduction

Switched-affine systems evolve according to one of a finite number of affine vector fields; in power electronics, the switch configuration selects the active field. Practical devices cannot switch arbitrarily fast, so every mode must stay active for at least a minimum dwell duration. Under such timing constraints the meaningful steady behavior is a periodic orbit rather than a static equilibrium [1, 2, 3].

A common architecture first picks a nominal mode sequence and switching schedule that generate a desired periodic orbit, then perturbs the internal switching instants to reject cycle-to-cycle deviations. Predictive formulations impose dwell-time constraints directly, but solve a constrained optimization at every update [4, 5, 3]. This work isolates the simpler law already implicit in the terminal linear-quadratic design: a static map from the cycle-to-cycle state error to switching-time perturbations.

The term *proportional* needs care here. The controller is not a scalar P controller and does not manipulate voltage, current, or duty ratio. It is a multi-input state-feedback law whose manipulated variables are signed displacements of switching instants, and its output passes through uniform dwell-time conditioning before being applied.

This draft contributes:

1. a derivation of the cycle-to-cycle proportional switching-time controller from the linearized periodic dynamics;
2. an explicit conditioning rule that enforces the minimum dwell duration without an online optimization problem;
3. a polyhedral characterization of the raw-action admissible region and its maximal positively invariant subset;
4. a common-Lyapunov certificate showing that the conditioner cannot destabilize the local closed loop, the condition under which that certificate fails, and a numerical check of its Lyapunov function against the nonlinear trajectory; and
5. a reproducible simulation study on a three-cell converter.

2 Problem Statement

2.1 System Dynamics and Sequence

Consider the switched-affine plant

$$\dot{\mathbf{x}}(t) = A_{\sigma(t)}\mathbf{x}(t) + \mathbf{b}_{\sigma(t)}, \quad \mathbf{x}(t) \in \mathbb{R}^n, \quad (1)$$

where $\sigma(t)$ selects one of M operating modes. Each cycle applies a fixed mode sequence $\boldsymbol{\Omega} = (\omega_1, \dots, \omega_N)$ at the nominal switching instants

$$0 = \bar{t}_0 < \bar{t}_1 < \dots < \bar{t}_N = T_p, \quad (2)$$

whose nominal dwell durations are $\Delta\bar{t}_i = \bar{t}_i - \bar{t}_{i-1}$. Perturbed instants and durations are written t_i and $\Delta t_i = t_i - t_{i-1}$.

The nominal schedule generates a target periodic orbit $\bar{\mathbf{x}}(t)$ with $\bar{\mathbf{x}}(T_p) = \bar{\mathbf{x}}(0)$, whose cycle-boundary state is $\mathbf{x}^* = \bar{\mathbf{x}}(0)$. This point is not the desired average operating point about which the orbit oscillates.

For mode ω_i , introduce the augmented state and mode matrix

$$\mathcal{X}(t) = \begin{bmatrix} \mathbf{x}(t) \\ 1 \end{bmatrix}, \quad F_i = \tilde{A}_{\omega_i} = \begin{bmatrix} A_{\omega_i} & \mathbf{b}_{\omega_i} \\ \mathbf{0}^\top & 0 \end{bmatrix}. \quad (3)$$

Exact propagation over a dwell duration Δt_i is then

$$\mathcal{X}(t_i) = \exp(F_i \Delta t_i) \mathcal{X}(t_{i-1}). \quad (4)$$

The simulation uses this expression as well, which removes numerical integration error within each constant-mode segment.

2.2 Linearization Procedure

Two distinct sets of decision variables appear below, and they must not be conflated. The *switching-instant offsets* $\delta\tau_{k,i}$ displace the interior instants of cycle k ,

$$t_{k,i} = \bar{t}_i + \delta\tau_{k,i}, \quad i = 1, \dots, N-1, \quad (5)$$

with fixed cycle boundaries $\delta\tau_{k,0} = \delta\tau_{k,N} = 0$. The *duration perturbations* $\delta t_{k,i} = \Delta t_{k,i} - \Delta \bar{t}_i$ follow from them as

$$\delta t_{k,i} = \delta\tau_{k,i} - \delta\tau_{k,i-1}, \quad i = 1, \dots, N. \quad (6)$$

There are N duration perturbations but only $N - 1$ independent offsets, because holding the period fixed forces $\sum_i \delta t_{k,i} = 0$.

Let $\mathbf{e}_k = \mathbf{x}_k - \mathbf{x}^*$ be the cycle-to-cycle error and let the nominal augmented transition of interval i be

$$\phi_i = \exp(F_i \Delta \bar{t}_i). \quad (7)$$

The augmented state at the start of cycle k is

$$\mathcal{X}_k = \bar{\mathcal{X}}(0) + \begin{bmatrix} \mathbf{e}_k \\ 0 \end{bmatrix}. \quad (8)$$

The exact one-cycle propagation under the perturbed durations is then

$$\mathcal{X}_{k+1} = e^{F_N(\Delta \bar{t}_N + \delta t_{k,N})} \dots e^{F_1(\Delta \bar{t}_1 + \delta t_{k,1})} \left(\bar{\mathcal{X}}(0) + \begin{bmatrix} \mathbf{e}_k \\ 0 \end{bmatrix} \right). \quad (9)$$

For one interval, the matrix exponential has the first-order expansion

$$e^{F_i(\Delta \bar{t}_i + \delta t_{k,i})} = \phi_i + F_i \phi_i \delta t_{k,i} + \mathcal{O}(\delta t_{k,i}^2). \quad (10)$$

Substituting it into (9) and keeping at most one perturbed factor per term gives

$$\begin{aligned} \mathcal{X}_{k+1} &\doteq \phi_N \dots \phi_1 \bar{\mathcal{X}}(0) + \phi_N \dots \phi_1 \begin{bmatrix} \mathbf{e}_k \\ 0 \end{bmatrix} \\ &\quad + \sum_{i=1}^N \phi_N \dots \phi_{i+1} F_i \phi_i \dots \phi_1 \bar{\mathcal{X}}(0) \delta t_{k,i}. \end{aligned} \quad (11)$$

Here $\doteq$ denotes equality to first order: terms with two timing perturbations, or with a timing perturbation multiplied by $\mathbf{e}_k$, are discarded. The rightmost factor acts on the initial state first, and an empty product is the identity.

Periodicity of the nominal trajectory gives $\phi_N \dots \phi_1 \bar{\mathcal{X}}(0) = \bar{\mathcal{X}}(0)$, so the nominal term on the right of (11) cancels $\bar{\mathcal{X}}(0)$ on the left. Writing the augmented error as $\tilde{\mathbf{e}}_k = \mathcal{X}_k - \bar{\mathcal{X}}(0)$ leaves the augmented duration-coordinate model [5, 3]

$$\tilde{\mathbf{e}}_{k+1} \doteq \tilde{\Phi} \tilde{\mathbf{e}}_k + \tilde{\Gamma} \delta \mathbf{t}_k, \quad (12)$$

where

$$\tilde{\Phi} = \phi_N \dots \phi_1, \quad (13)$$

$$\tilde{\Gamma} = [\tilde{\mathbf{g}}_1 \quad \dots \quad \tilde{\mathbf{g}}_N], \quad \tilde{\mathbf{g}}_i = \phi_N \dots \phi_{i+1} F_i \phi_i \dots \phi_1 \bar{\mathcal{X}}(0). \quad (14)$$

Equation (12) completes the linearization. Every quantity in it is an $(n + 1)$ -dimensional product of nominal transitions, and no coordinate change has been made yet.

2.3 Projection to Physical Coordinates

The augmented model carries one redundant coordinate. Removing it is a change of representation rather than a further approximation, and it is what makes the controller design of section 3 an n -dimensional problem.

By (3) each F_i is block upper triangular with zero last row, so each nominal transition is

$$\phi_i = \begin{bmatrix} e^{A_{\omega_i} \Delta \bar{t}_i} & \mathbf{h}_i \\ \mathbf{0}^\top & 1 \end{bmatrix}, \quad \mathbf{h}_i = \int_0^{\Delta \bar{t}_i} e^{A_{\omega_i} s} \mathrm{d}s \, \mathbf{b}_{\omega_i}, \quad (15)$$

where $\mathbf{h}_i$ is the particular-solution term that a non-augmented derivation would carry separately. Products of such matrices keep this structure, so $\tilde{\Phi}$ also has last row $[\mathbf{0}^\top \, 1]$.

Two consequences follow. First, $\mathcal{X}_k$ and $\bar{\mathcal{X}}(0)$ share the trailing entry 1, so the augmented error is $\tilde{\mathbf{e}}_k = [\mathbf{e}_k^\top \, 0]^\top$. Second, F_i has zero last row, so each $\tilde{\mathbf{g}}_i$ likewise ends in zero. The last row of (12) is therefore the trivial identity $0 = 0$, and discarding it loses nothing.

Introduce the projection $P = [I_n \, \mathbf{0}] \in \mathbb{R}^{n \times (n+1)}$, so that $\mathbf{e}_k = P\tilde{\mathbf{e}}_k$ and $\tilde{\mathbf{e}}_k = P^\top \mathbf{e}_k$. Left-multiplying (12) by P gives the duration-coordinate model in physical coordinates,

$$\mathbf{e}_{k+1} \doteq \Phi \mathbf{e}_k + \Gamma \delta \mathbf{t}_k, \quad \Gamma = [\mathbf{g}_1 \quad \cdots \quad \mathbf{g}_N], \quad (16)$$

with

$$\Phi = P\tilde{\Phi}P^\top = e^{A_{\omega_N} \Delta \bar{t}_N} \cdots e^{A_{\omega_1} \Delta \bar{t}_1}, \quad (17)$$

$$\mathbf{g}_i = P\tilde{\mathbf{g}}_i. \quad (18)$$

The affine columns $\mathbf{h}_i$ do not appear in Φ , because in (15) they multiply the zero trailing entry of $\tilde{\mathbf{e}}_k$. They still act on Γ , through the nominal orbit $\bar{\mathcal{X}}(0)$ in (14).

2.4 Switching-Instant Coordinates

Substituting (6) converts (16) into the fixed-period switching-instant model

$$\mathbf{e}_{k+1} \doteq \Phi \mathbf{e}_k + \Gamma_\tau \delta \boldsymbol{\tau}_k, \quad \Gamma_\tau = [\mathbf{g}_1 - \mathbf{g}_2 \quad \cdots \quad \mathbf{g}_{N-1} - \mathbf{g}_N], \quad (19)$$

with $\Gamma_\tau \in \mathbb{R}^{n \times (N-1)}$. Equivalently $\Gamma_\tau = \Gamma D$, where D is the difference map defined in (29). The sensitivity to internal switching instant i is

$$\Gamma_{\tau,i} = P\phi_N \cdots \phi_{i+1} (F_i - F_{i+1}) \bar{\mathcal{X}}(\bar{t}_i), \quad i = 1, \dots, N-1, \quad (20)$$

which follows from $F_{i+1}\phi_{i+1} = \phi_{i+1}F_{i+1}$. The augmented duration linearization thus recovers the vector-field jump at each internal switch while preserving the cycle period. Equation (19) is local: its accuracy degrades as the state and switching schedule move away from the nominal orbit.

3 Proportional Switching-Time Controller

3.1 Cost Function and Linear-Quadratic Gain

The raw control law is the static state feedback

$$\delta \boldsymbol{\tau}_k^{\text{raw}} = -K \mathbf{e}_k. \quad (21)$$

The gain solves the discrete-time linear-quadratic regulator problem for (Φ, Γ_τ) ,

$$K = \text{dlqr}(\Phi, \Gamma_\tau, Q, R_\tau), \quad (22)$$

minimizing the infinite-horizon criterion

$$\sum_{k=0}^{\infty} \left(\mathbf{e}_k^\top Q \mathbf{e}_k + \delta \boldsymbol{\tau}_k^\top R_\tau \delta \boldsymbol{\tau}_k \right). \quad (23)$$

The subscript on R_τ distinguishes the timing weight from the load resistance R of section 4. Because $\delta \boldsymbol{\tau}_k$ carries units of time, the relative scaling of Q and R_τ depends on the time unit chosen for Γ_τ ; section 5 returns to this point.

Proportional therefore denotes a memoryless linear relation between the sampled state error and the switching-time perturbation vector, with a matrix gain synthesized from the local one-cycle model rather than tuned as a scalar.

3.2 Dwell-Time Constraints

A candidate perturbation produces the dwell duration

$$\Delta t_{k,i} = \Delta \bar{t}_i + \delta \tau_{k,i} - \delta \tau_{k,i-1}. \quad (24)$$

The hardware constraint is

$$\Delta t_{k,i} \geq \Delta t_{\min}, \quad i = 1, \dots, N. \quad (25)$$

Componentwise clipping of $\delta \boldsymbol{\tau}_k^{\text{raw}}$ does not enforce (25), because each internal switching instant affects two adjacent dwell durations. Instead, let

$$\delta t_{k,i}^{\text{raw}} = \delta \tau_{k,i}^{\text{raw}} - \delta \tau_{k,i-1}^{\text{raw}} \quad (26)$$

be the duration perturbation requested by the raw law. A single conditioning factor $\beta_k \in [0, 1]$ is selected as

$$\beta_k = \min \left\{ 1, \min_{i: \delta t_{k,i}^{\text{raw}} < 0} \frac{\Delta t_{\min} - \Delta \bar{t}_i}{\delta t_{k,i}^{\text{raw}}} \right\}, \quad (27)$$

where the inner minimum is dropped when no interval shrinks. The applied perturbation is

$$\delta \boldsymbol{\tau}_k = \beta_k \delta \boldsymbol{\tau}_k^{\text{raw}}. \quad (28)$$

Because every nominal interval satisfies $\Delta \bar{t}_i \geq \Delta t_{\min}$, the ratios in (27) are nonnegative and the resulting schedule satisfies (25). Conditioning preserves the direction of the raw LQR action and sacrifices only its magnitude, and it does so whenever a dwell constraint becomes active.

The online work is one matrix–vector product, a set of interval differences, scalar comparisons, and one vector scaling; no quadratic program is solved. That simplicity is the main advantage over constrained predictive control. The cost is conservatism: a single critical interval attenuates every switching-time perturbation.

3.3 Constraint-Admissible and Invariant Regions

Let $D \in \mathbb{R}^{N \times (N-1)}$ map switching-instant offsets to duration perturbations,

$$\delta \mathbf{t}_k = D \delta \boldsymbol{\tau}_k, \quad D = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ -1 & 1 & \ddots & \vdots \\ 0 & \ddots & \ddots & 0 \\ \vdots & \ddots & -1 & 1 \\ 0 & \cdots & 0 & -1 \end{bmatrix}. \quad (29)$$

Under the raw law (21), the dwell constraints hold without conditioning when

$$\Delta \bar{\mathbf{t}} - DK \mathbf{e}_k \geq \Delta t_{\min} \mathbf{1}_N. \quad (30)$$

The raw-action admissible region is therefore the polyhedron

$$\mathcal{C} = \{\mathbf{e} \in \mathbb{R}^n : DK \mathbf{e} \leq \Delta \bar{\mathbf{t}} - \Delta t_{\min} \mathbf{1}_N\}. \quad (31)$$

Inside $\mathcal{C}$, $\beta_k = 1$ and the LQR action passes through unchanged. Outside $\mathcal{C}$ the controller never becomes infeasible: $\beta_k = 0$ recovers the nominal schedule, and (27) selects the largest scaling permitted along the raw-action direction.

Let the unconditioned local closed-loop matrix be

$$A_{\text{cl}} = \Phi - \Gamma_{\tau} K. \quad (32)$$

The maximal positively invariant subset of $\mathcal{C}$ is

$$\mathcal{C}_{\infty} = \bigcap_{\ell=0}^{\infty} \left\{ \mathbf{e} \in \mathbb{R}^n : A_{\text{cl}}^{\ell} \mathbf{e} \in \mathcal{C} \right\}. \quad (33)$$

If A_{cl} is Schur stable, every initial error in $\mathcal{C}_{\infty}$ stays raw-action admissible and converges to the origin under the local linear model without ever activating the conditioner. This is a linear-model certificate, not a region-of-attraction guarantee for the nonlinear switched plant.

3.4 Stability Under Active Conditioning

Inside $\mathcal{C}_{\infty}$ the conditioner never acts, so (33) settles the question. Outside it the applied gain is $\beta_k K$ with $\beta_k < 1$, and neither the LQR construction nor the invariance argument applies. This subsection closes that gap.

Substituting the conditioned law (28) into (19) gives

$$\mathbf{e}_{k+1} \doteq A(\beta_k) \mathbf{e}_k, \quad A(\beta) = \Phi - \beta \Gamma_{\tau} K, \quad (34)$$

and since $A_{\text{cl}} = \Phi - \Gamma_{\tau} K$,

$$A(\beta) = (1 - \beta) \Phi + \beta A_{\text{cl}}. \quad (35)$$

The conditioned closed loop is therefore a convex combination of the open-loop monodromy matrix Φ , recovered at $\beta = 0$ where the nominal schedule is applied unchanged, and the unconditioned design A_{cl} , recovered at $\beta = 1$. Conditioning cannot move the closed loop off the segment joining these two matrices.

Because β_k depends on $\mathbf{e}_k$ through (27), (34) is nonlinear. The following certificate sidesteps that dependence by covering every admissible scaling sequence at once.

Proposition 1. *Suppose there exists $P \succ 0$ with*

$$\Phi^\top P \Phi - P \prec 0, \quad A_{\text{cl}}^\top P A_{\text{cl}} - P \prec 0. \quad (36)$$

Then $V(\mathbf{e}) = \mathbf{e}^\top P \mathbf{e}$ is a common Lyapunov function for (34) for every sequence $\{\beta_k\} \subset [0, 1]$, and the origin of the local model is globally exponentially stable.

Proof. Let $\beta \in [0, 1]$. Using $\Phi - A_{\text{cl}} = \Gamma_\tau K$ and (35),

$$\begin{aligned} (1 - \beta)\Phi^\top P \Phi + \beta A_{\text{cl}}^\top P A_{\text{cl}} - A(\beta)^\top P A(\beta) \\ = \beta(1 - \beta) K^\top \Gamma_\tau^\top P \Gamma_\tau K \succeq 0, \end{aligned} \quad (37)$$

because $P \succ 0$ and $\beta(1 - \beta) \geq 0$. Hence $A(\beta)^\top P A(\beta) \preceq (1 - \beta)\Phi^\top P \Phi + \beta A_{\text{cl}}^\top P A_{\text{cl}} \prec P$ by (36), so V decreases uniformly along the whole segment. $\square$

Two consequences deserve emphasis. First, evaluating (36) at $\beta = 0$ requires Φ to be Schur, so the certificate exists only when the nominal orbit is already open-loop stable. If $\rho(\Phi) > 1$, then $\beta_k = 0$ reproduces an unstable schedule, deep conditioning is genuinely unsafe, and the regional analysis of section 3.3 becomes the only guarantee available. Second, the worst case over the segment is $\beta = 0$, so the rate certified by (36) is the open-loop rate. Proposition 1 establishes that conditioning cannot destabilize the local loop; it promises nothing about how fast convergence is while the conditioner is active.

4 Case Study

4.1 Multilevel Converter

The plant is a three-cell multilevel DC–DC converter with state $\mathbf{x} = [v_{C_1}, v_{C_2}, i_L]^\top$. For binary switch commands $u_1, u_2, u_3 \in \{0, 1\}$, the mode-dependent dynamics are

$$A_{\mathbf{u}} = \begin{bmatrix} 0 & 0 & (u_2 - u_1)/C_1 \\ 0 & 0 & (u_3 - u_2)/C_2 \\ (u_1 - u_2)/L & (u_2 - u_3)/L & -R/L \end{bmatrix}, \quad \mathbf{b}_{\mathbf{u}} = \begin{bmatrix} 0 \\ 0 \\ Eu_3/L \end{bmatrix}. \quad (38)$$

The model therefore has $M = 8$ operating modes. Table 1 lists the simulation parameters.

Table 1: Simulation parameters of the PATINO_2 configuration.

Quantity	Symbol	Value
Source voltage	E	30 V
Capacitances	C_1, C_2	40 μF
Inductance	L	10 mH
Load resistance	R	10 Ω
Nominal period	T_p	286 μs
Minimum dwell duration	$\Delta t_{\min}$	3 μs
State weight	Q	I_3
Timing weight	R_τ	I_8 (seconds ⁻²)

The cycle uses $N = 9$ mode intervals, so the controller manipulates $N - 1 = 8$ internal switching instants. The one-based dynamics indices are

$$\mathbf{\Omega} = (1, 2, 4, 8, 3, 1, 5, 8, 5), \quad (39)$$

which correspond to the physical mode identifiers $\mathbf{\Omega} - 1 = (0, 1, 3, 7, 2, 0, 4, 7, 4)$. The nominal schedule in microseconds is

$$\bar{\mathbf{t}} = (0, 66, 88, 110, 132, 154, 220, 242, 264, 286), \quad (40)$$

giving nominal dwell durations $\Delta \bar{t}_i$ of 66 or 22 μs , both far above $\Delta t_{\min}$. The cycle-boundary state of the target orbit is

$$\mathbf{x}^* = [9.9247, 19.2928, 0.9823]^\top, \quad (41)$$

while the desired average operating point $[10, 20, 1]^\top$ is used to design the orbit but is never the feedback target.

These values differ from the earlier thesis snapshot, which reports $[9.5143, 19.8211, 1.0314]^\top$ and a slightly different schedule [3]; this article uses the current configuration throughout.

4.2 Simulation Setup

The exact segment propagation (4) runs for 80 cycles from the initial state

$$\mathbf{x}_0 = [7.51, 20.82, 0.03]^\top. \quad (42)$$

The controlled and open-loop runs share this initial state and the nominal mode sequence. Because the three states carry different units and magnitudes, the reported metric is normalized:

$$\varepsilon_k = \|\text{diag}(10, 20, 1)^{-1}(\mathbf{x}_k - \mathbf{x}^*)\|_2. \quad (43)$$

The script `scripts/generate_results.m` runs the experiment and emits every scalar quoted below to `results/metrics.tex`. It also builds $\mathcal{C}$ from (31), computes $\mathcal{C}_\infty$ by invariant-set iteration, solves the vertex conditions (36) for P , sweeps β to bound $\|A(\beta)\|_P$, and exports both vertex and half-space representations together with A_{cl} . The claims of section 3.4 are enforced as runtime assertions, so a configuration that violated them would fail the build rather than be reported.

5 Results

Figure 1 compares the continuous state trajectories. The controlled response reaches a neighborhood of the target orbit within a few cycles; the open-loop response settles on a different cycle-boundary behavior over the same interval.

The normalized error in figure 2 falls from 0.9854 to 0.0086 over 80 cycles, a 99.1% reduction. The open-loop error ends at 0.2344, a reduction of only 76.2%. The nonzero controlled residual reflects the mismatch between the nonlinear periodic plant and the local first-order model, together with the early activation of the timing conditioner.

Figure 3 follows the conditioner through the first 12 cycles, which is where all of the timing activity occurs. The four panels separate what the controller asked for, what it was allowed to apply, the scalar that reconciles the two, and the constraint being protected.

The raw law requests a largest offset of 93.1 μs , more than four times the 22 μs nominal dwell durations that the corresponding instants bound. That schedule is not merely aggressive but grossly infeasible: applied unmodified it would drive several durations negative. Conditioning reduces it to 19.0 μs , scaling the action to as little as $\beta_k = 0.204$ of its raw magnitude. The shortest dwell duration observed over the complete run is 3.0 μs , exactly the prescribed lower bound and never below it.

The bottom two panels show why the bound is met exactly rather than with margin. While $\beta_k < 1$ the conditioner scales until the most critical duration sits precisely on $\Delta t_{\min}$, so the constraint

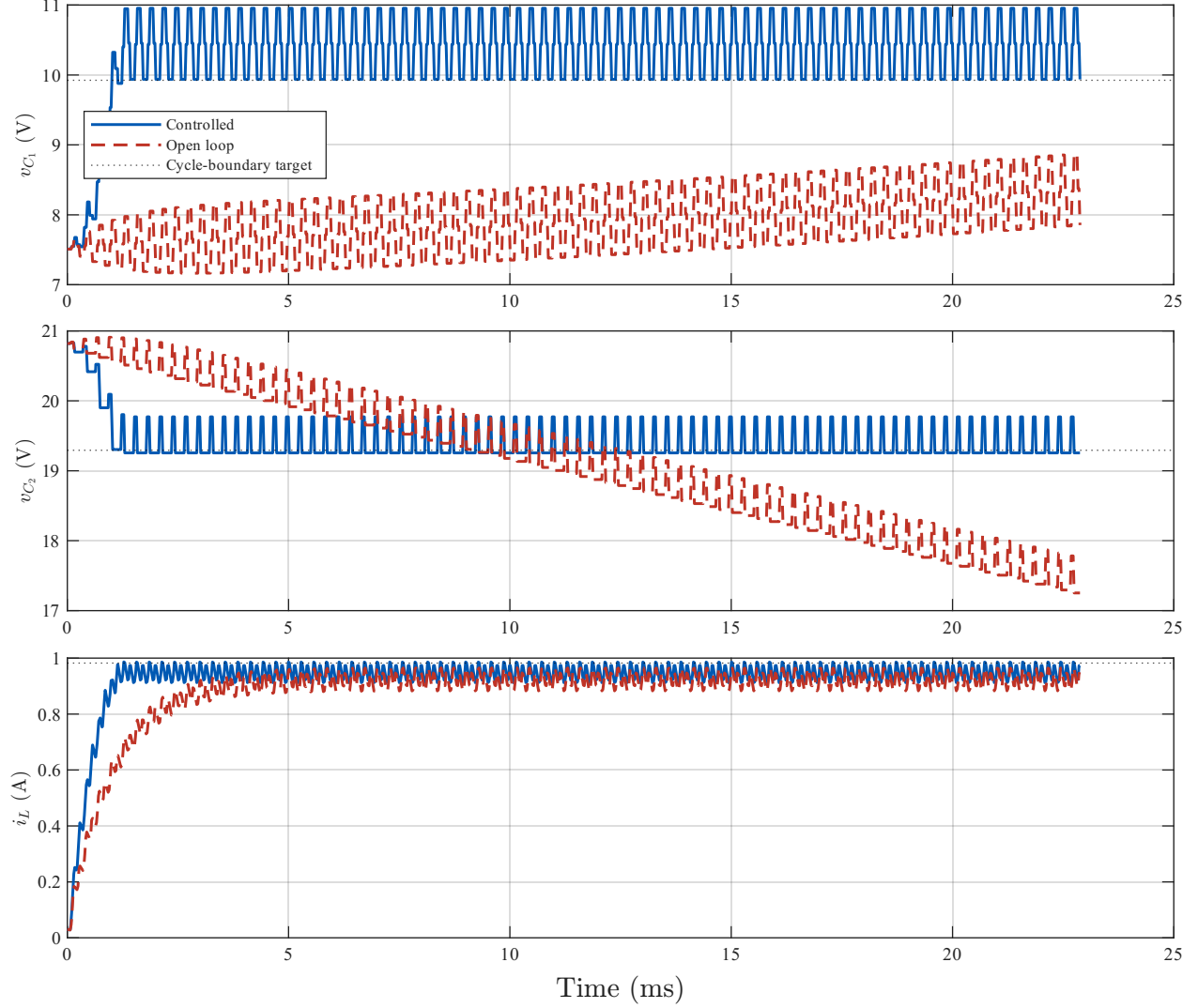


Figure 1: State response with proportional switching-time control and in open loop. The dotted values mark the cycle-boundary target, not the complete periodic waveform.

is active by construction on every conditioned cycle. Once the trajectory enters $\mathcal{C}$ at cycle 4, β_k returns to one, the raw and applied actions coincide, and the shortest duration lifts clear of the bound. The applied offsets remain comparable to the nominal durations throughout the transient, confirming that these early cycles operate well outside the region where (19) is accurate.

The computed gain gives a local unconditioned closed loop with spectral radius $\rho(A_{cl}) = 5.86 \times 10^{-8}$. This value is numerically zero rather than merely stable, and the reason is the weighting in (23): with $\delta\tau_k$ expressed in seconds, $\delta\tau_k^T R_\tau \delta\tau_k$ is of order 10^{-9} while $e_k^T Q e_k$ is of order one, so $R_\tau = I_8$ is effectively no penalty at all. Since Γ_τ has eight columns and full row rank, the design collapses to the minimum-norm deadbeat gain. Reporting R_τ without its time unit would therefore overstate how much the timing effort was actually weighted.

The invariant-set iteration returns $\mathcal{C}_\infty = \mathcal{C}$ to numerical tolerance, and the polyhedron is bounded and full-dimensional in the three error coordinates. Every error in the displayed volume therefore stays there under the local linear model while the raw LQR action remains dwell-admissible. This

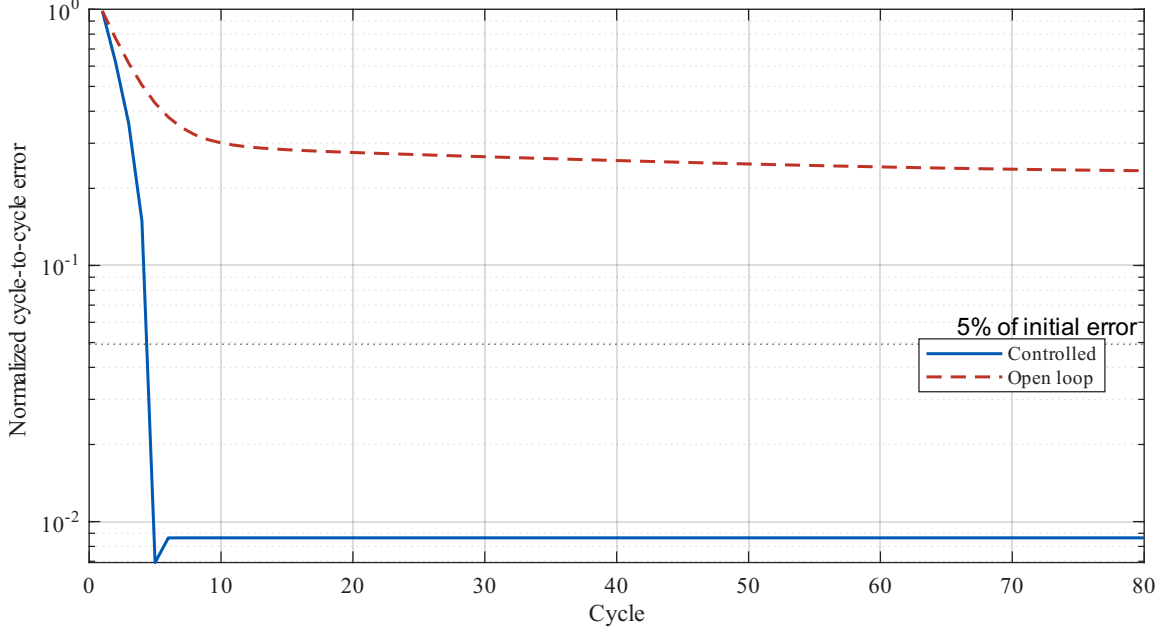


Figure 2: Normalized cycle-to-cycle error defined by (43).

coincidence is a direct consequence of the near-deadbeat gain: $A_{\text{cl}}\mathbf{e} \approx \mathbf{0}$ for every bounded $\mathbf{e}$, and the origin lies in the interior of $\mathcal{C}$ because $\bar{d}_i > d_{\min}$ for all i . The equality $\mathcal{C}_\infty = \mathcal{C}$ should not be read as a property of the conditioning scheme; a gain with a meaningful timing penalty would generally yield $\mathcal{C}_\infty \subsetneq \mathcal{C}$.

Figure 5 presents the region and trajectory in orthographic XY, XZ, and YZ projections together with a perspective XYZ view. The simulated initial error lies outside the set. Conditioning is active for the first 3 cycles; the trajectory enters $\mathcal{C}$ at cycle 4 and stays inside for the rest of the run.

Those first 3 cycles are exactly the regime that proposition 1 covers. The monodromy matrix has $\rho(\Phi) = 0.99986$, so (36) is feasible and P can be taken as the solution of $\Phi^T P \Phi - P = -I$; that same P satisfies the second inequality with an enormous margin, because $A_{\text{cl}} \approx 0$ makes $A_{\text{cl}}^T P A_{\text{cl}}$ numerically negligible against P . Sweeping β across $[0, 1]$ gives $\sup_\beta \|A(\beta)\|_P = 0.99989$, where $\|A\|_P$ is the matrix norm induced by $\|\mathbf{e}\|_P = (\mathbf{e}^T P \mathbf{e})^{1/2}$, and the supremum is attained at $\beta = 0$ as predicted.

The certified decay is therefore only a factor 0.99989 per cycle, which is no practical convergence bound at all, even though the simulated error falls by 99.1%. The gap is not slack in the analysis: $\rho(\Phi)$ is genuinely close to one because the converter's LC resonance is very lightly damped, so a fully conditioned cycle really does make almost no progress. Fast convergence in the simulation comes from β_k returning to one after cycle 4, not from contraction during conditioning. For this plant proposition 1 should be read as a safety statement, not a performance one.

Proposition 1 is a statement about the local model (19), so it is worth asking whether its Lyapunov function actually decreases along the nonlinear switched trajectory. Figure 4 evaluates $V(\mathbf{e}_k) = \mathbf{e}_k^T P \mathbf{e}_k$ on the simulated cycle errors using the same P . Over the 3 conditioned cycles the worst observed ratio is $V(\mathbf{e}_{k+1})/V(\mathbf{e}_k) = 0.8625$, comfortably below one, so the certificate is not merely satisfied but satisfied with margin in precisely the regime it was constructed for.

Monotonicity does, however, fail later in the run: the worst ratio over all 80 cycles is 1.0149, first exceeding one at cycle 6. This is not a counterexample to proposition 1, because by that cycle V has already fallen by a factor of 6525 and the trajectory has reached the residual floor set by the

mismatch between the nonlinear plant and the first-order model. What the increase measures is that floor, not divergence; the error stays bounded and the final normalized value is 0.0086. The practical reading is that the certificate governs the transient, where conditioning is active and the errors are large, and stops being informative once the trajectory reaches the accuracy limit of the linearization.

The state-space view in figure 6 emphasizes that the controller does not drive the continuous state to a static point. It aligns the cycle boundary with the selected phase of the target periodic orbit while the state keeps traversing that orbit within every switching cycle.

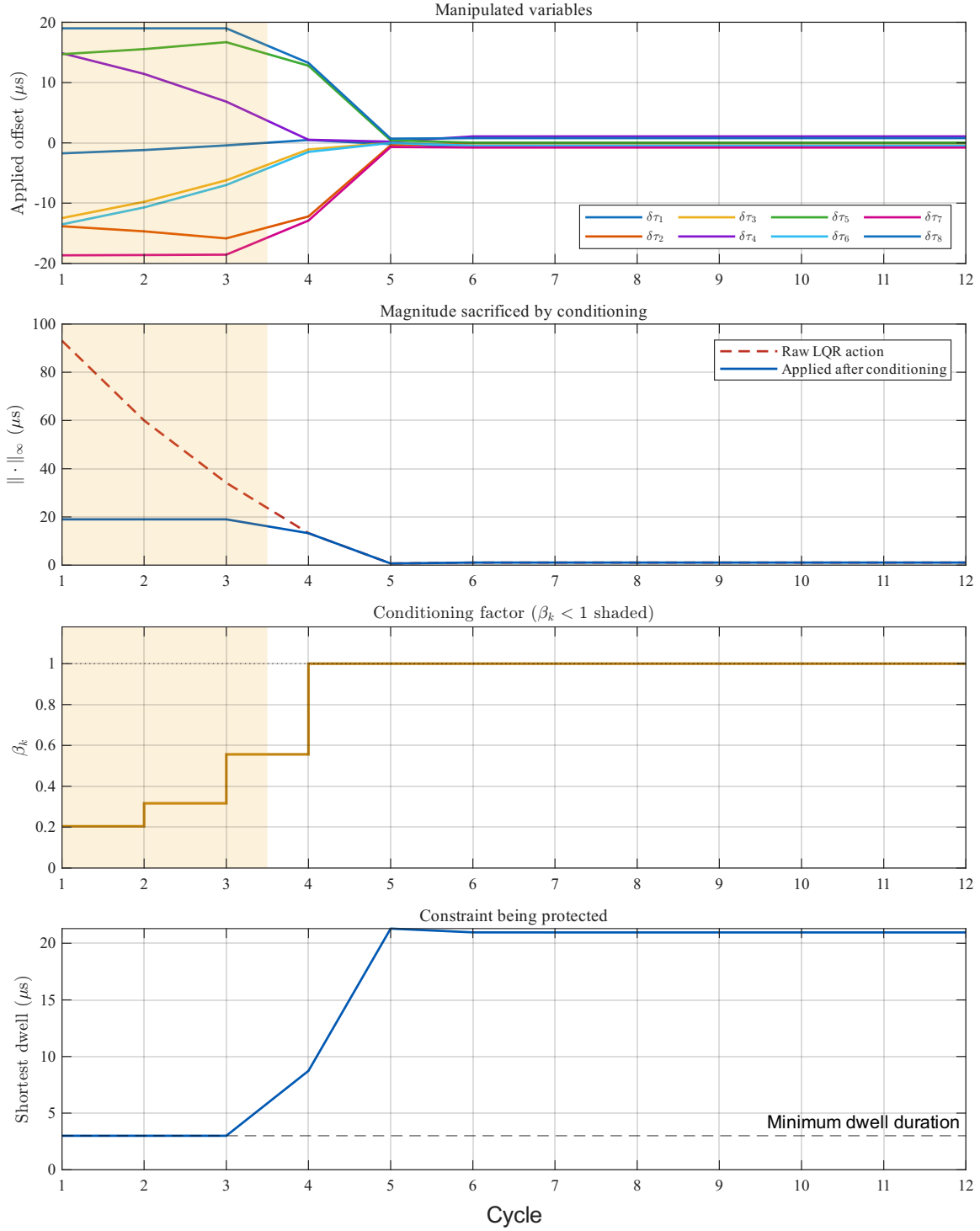


Figure 3: Dwell-time conditioning over the first 12 cycles. From top: the eight applied switching-instant offsets; the raw LQR action against the applied action in the ∞ -norm; the conditioning factor β_k ; and the shortest dwell duration against its lower bound. Shading marks the cycles on which $\beta_k < 1$. The raw action peaks at 93.1 μs , well beyond what the 22 μs nominal durations admit.

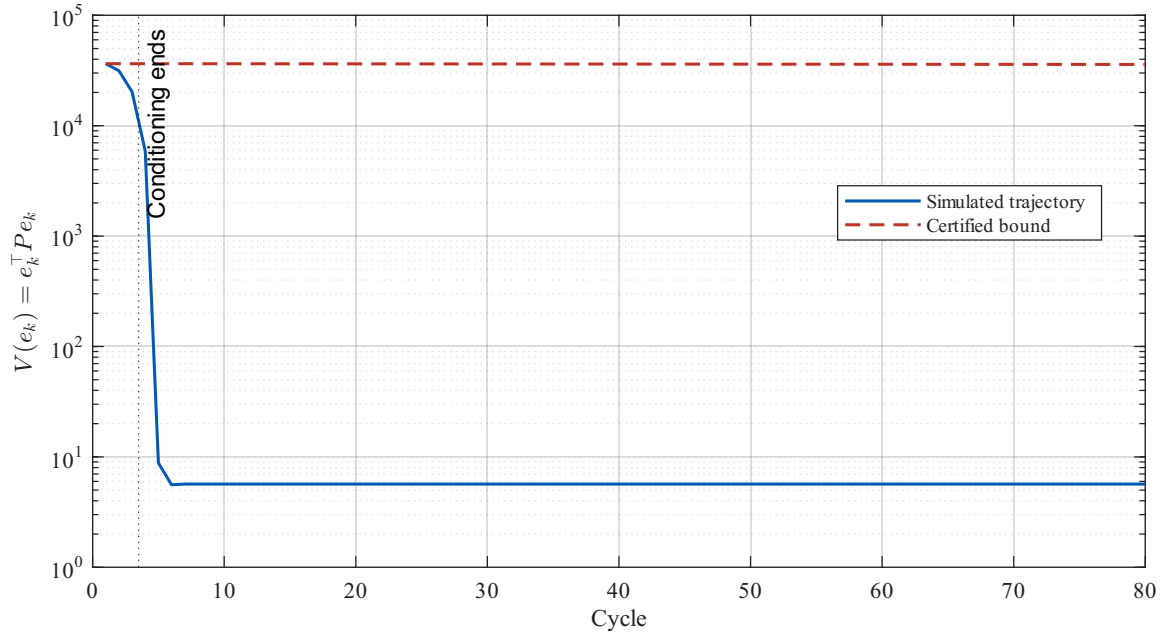


Figure 4: Lyapunov function $V(\mathbf{e}_k) = \mathbf{e}_k^T P \mathbf{e}_k$ of proposition 1 evaluated on the simulated trajectory, against the decay certified by $\sup_{\beta} \|A(\beta)\|_P$. The certified bound is far more conservative than the observed decay.

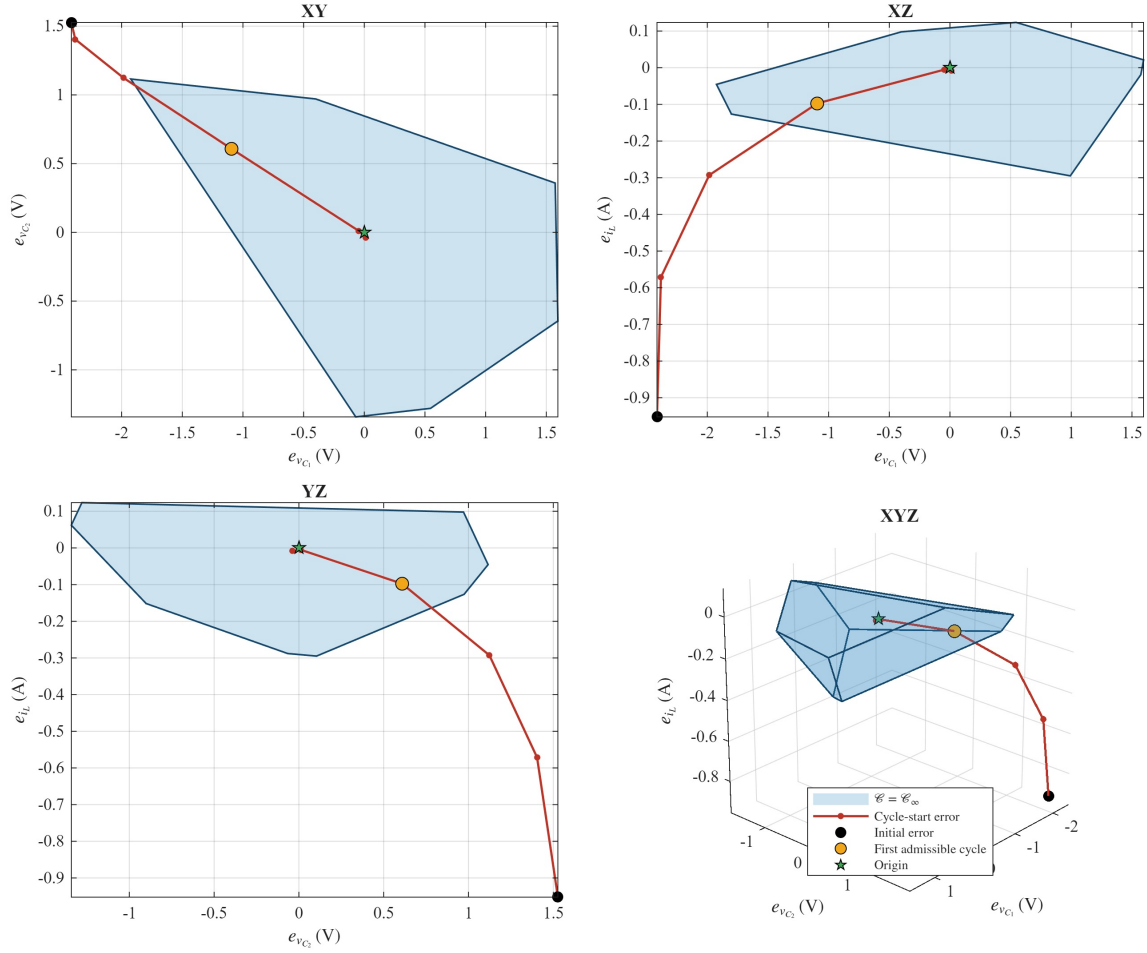


Figure 5: Orthographic and perspective views of the raw-action admissible region and the simulated cycle-start error trajectory.

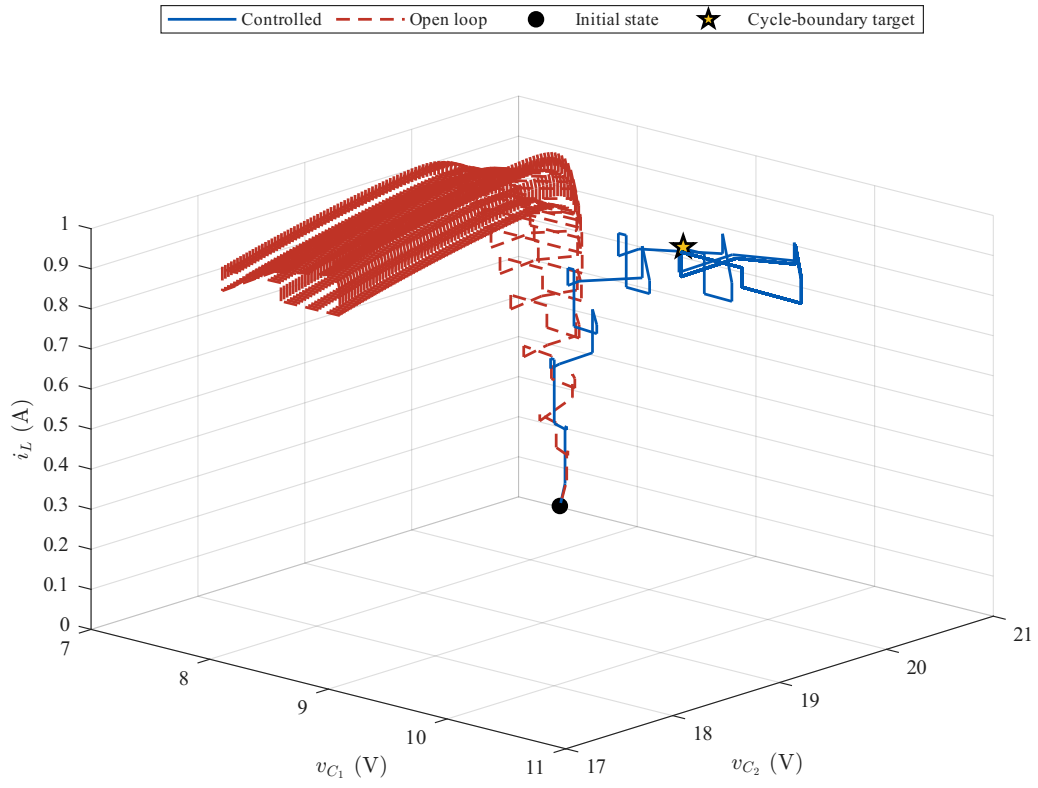


Figure 6: Controlled and open-loop trajectories in converter state space.

6 Discussion

The simulation supports proportional switching-time feedback as a low-cost baseline. Once K is computed offline, the online controller needs no prediction horizon and no optimization solver, and it states the actuator constraint directly in timing coordinates, where dwell-time feasibility is checked by inspection.

Five limitations qualify the result. First, the one-cycle model is a first-order approximation around a single phase of a single periodic orbit, so neither $\mathcal{C}_\infty$ nor proposition 1 guarantees convergence for the nonlinear plant; both are statements about (19). Second, the uniform factor in (27) is conservative: one critical dwell duration scales all perturbations. Third, conditioning alters the effective closed-loop gain, so the unconstrained LQR optimality claim lapses whenever $\beta_k < 1$, and while proposition 1 shows the loop stays stable, the rate it certifies collapses to the open-loop rate and is therefore vacuous for a lightly damped orbit such as this one. Fourth, the reported gain is effectively deadbeat because R_τ carries no weight at the second time scale; a design with a meaningful timing penalty would trade the aggressive first cycles, in which the linearization is least trustworthy, against slower convergence, and that trade-off is not explored here. Fifth, the evidence is deterministic simulation only: measurement noise, parameter uncertainty, computation delay, timer quantization, and switching losses lie outside the experiment.

The controller is therefore a constrained baseline and a candidate fallback law, not a replacement for predictive control across all operating regions. The natural next study holds the target orbit, model, initial conditions, and timing constraints fixed while comparing convergence, computation time, the raw-action admissible region, and the feasibility region of dwell-time-constrained MPC.

7 Conclusion

This paper formulated a proportional controller for periodic-orbit stabilization of switched-affine systems directly in switching-time coordinates, combining a discrete LQR gain on the local one-cycle error model with a scalar conditioning operation that enforces the minimum dwell duration. The raw-action admissible polyhedron is the counterpart of an MPC feasibility analysis: inside it conditioning is inactive, and its invariant subset certifies convergence under the unmodified gain in the local linear model. The two regions coincide for the present configuration, a consequence of the near-deadbeat gain rather than of the conditioning scheme. Outside that region the conditioned loop is a convex combination of the open-loop monodromy matrix and the unconditioned design, so a single quadratic Lyapunov function certifies that conditioning cannot destabilize the local model whenever the nominal orbit is itself open-loop stable. In the three-cell converter simulation the controller cut the normalized cycle-to-cycle error by 99.1% over 80 cycles, held the $3\text{ }\mu\text{s}$ dwell-time bound, and entered the admissible region after 3 conditioned cycles. The approach is computationally simple and physically interpretable, but it stays local and grows conservative whenever the uniform conditioner is active. Experimental validation and a controlled comparison with constrained MPC are needed before stronger robustness or real-time claims are warranted.

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